Marvell: Does the Renewed AWS Partnership Cement Marvell as a PCI Alternative to Astera?

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Viewpoint: Former Executive | Moderator: Sheetal Duggal (SD) | Dec 05, 2024 | 41 Min Read

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SUMMARY

Overview

The discussion, led by Sheetal Duggal from Guidepoint Insights with Raj Singh, a former EVP at Marvell, focuses on the implications of the expanded partnership between Marvell and AWS on the PCIe and switch market. The conversation delves into how this partnership affects other players like Astera Labs and the broader competitive landscape in the semiconductor industry.

Key Insights

- **AWS and Marvell Partnership**: The partnership is not limited to ASIC supply but includes a broader range of products and services, indicating a deep strategic alignment.
- **Astera Labs' Position**: Astera's heavy reliance on AWS for PCIe retimers poses a risk, especially with technological shifts and increased competition.
- **Technological Shifts**: The market is experiencing a shift from PCIe to faster Ethernet solutions, driven by the need for higher data transfer speeds in data centers.

Competitive Landscape

Company	Strengths	Weaknesses/Threats
Marvell	Strong AWS partnership, diverse product offerings	Competition from Broadcom, technological shifts
Astera Labs	Early mover advantage in PCIe retimers	Heavy reliance on AWS, technological shifts
Broadcom	Established market presence	Customer dissatisfaction, competition

Applications & Use Cases

- **PCIe Retimers and Switches**: Used for data signal integrity in data centers, but facing challenges due to the shift towards optical and faster Ethernet solutions.
- **ASICs**: Increasingly important for custom solutions in hyperscaler environments, with AWS and other hyperscalers investing heavily in custom ASIC development.

Quantitative Insights

Metric	Value/Insight
Marvell's P&L (2023)	$1.8 billion to $1.9 billion
AWS Warrants	4 million warrants to buy Marvell stock
Astera's Revenue Source	Majority from PCIe retimers, heavily AWS-dependent

Market Strategy

- **Marvell**: Leveraging strategic partnerships with AWS to secure long-term market positioning and access to new technologies.
- **Astera Labs**: Needs to diversify its product offerings and customer base to mitigate risks associated with heavy reliance on AWS.

Stage & Timing

- **PCIe Market**: Currently in a transitional phase with a shift towards faster Ethernet solutions expected within the next 12-24 months.
- **Marvell and AWS Partnership**: Set for a five-year term, indicating a long-term strategic alignment.

The discussion highlights the dynamic nature of the semiconductor market, with technological advancements driving shifts in competitive positioning and market strategies.

SD: Good morning. Sheetal Duggal at Guidepoint Insights. Thanks for dialing into our call today. We're going to be talking about Marvell and PCIe as well as switch market, and how the AWS partnership affects it, what the implications are for other players such as Astera Labs. We have Raj Singh here to talk through his thoughts on the topic. As always, if you have any questions, please email them to ask@guidepoint.com.

I'll work through any questions I receive on anonymous basis as they come in. The email is just again is A-S-K @guidepoint.com. With that, Raj, it's a real pleasure to have you here. Thank you so much for the time. Would love to have you walk through your background for the audience, how you kind of interacted with the market, experience with Astera, and then we can kind of dive into the competitive landscape and kind of where you think things are going for the space.

RS: Certainly. It's a great pleasure to be here. Thank you for having me. So I retired from Marvell about a year-and-a-half ago in May of 2023. And my last transitional role was EVP of compute strategy and business development, specifically focused on the hyperscaler market. Prior to that, I ran what was then and it's still now the single largest business group inside the company, which was the process of business group, and it's now called custom compute and storage because they folded storage into it as well.

And that's at the time, about a $1.8 billion, $1.9 billion P&L, about 2,000 people, of course with some additions now and with the rise of the hyperscaler market for them, it's got somewhat bigger. Prior to Marvell, I came to Marvell through the acquisition of Cavium, and I ran the processor business and the Ethernet business there as well, as well as the 5G business, which I started from scratch in 2012. And I've been in semiconductors for 40-something years. So that's my background. Happy to be here.

SD: [INAUDIBLE 2:25-2:38].

RS: Hello? Hello? Hello? Are you there? Hello? Is somebody speaking? I can't hear them. It's very muffled.

SD: Raj, can you hear me? Raj? Raj, can you hear me? Apologies to the audience. I think we're having technical difficulty. We will be back. Apologies to the audience. We'll resume in a minute.

RS: Hello?

SD: Hi, Raj. How are you? Sorry about that.

RS: Oh, I don't know what happened. But I just got locked up. Yes, I can hear you fine.

SD: Yes. Got it. So just to recap, what I wanted to unpack a little bit was some of the announcement that came out of AWS, obviously the expanded AWS Marvell partnership, where Marvell offered AWS, I think it was four million or so warrants to buy Marvell stock, which felt unusual for a public company, but not - unsurprising.

And then really, you know, obviously, some of the announcements that re:Invent talking about, you know, where AWS is going, how they're thinking about ASICs, and what this really means for the competitive landscape when we think about Astera, Marvell, even Broadcom, and the various players?

RS: Yes. Ok. So let's do this portion by portion. The AWS announcement was - I was trying to remember how long ago this conversation started. I think over four years ago when we first started that - initiated that conversation, it's taken its twists and turns, and I'm glad they finally got it over the line.

But it's been going on for - it's not a thing that happened yesterday, and it's a long term desire on both parties to come to in understanding that would benefit both of them by giving AWS access to certain technologies and giving Marvell access to certain parts of the market.

You should think of it not just as an ASIC supply agreement, but as a much broader agreement, which covers standard products, some sorts of services, semi-custom products as well as full custom products. So it's much broader than, hey, let me do three sets of ASICs for you over five years, which is if you read it in - at face value, that's what you get.

But then talk about - and this is all public, of course, then talk about, oh, agreement also covers other things like optics, and retimers, and things. And I think there's your clue that it's a much broader agreement than you would look at it at face value.

SD: Yes, that's really helpful. So I guess when you think about that - and obviously, yes, the press release covered - it had to be five or six categories on top of ASICs. So it was a very - it was relatively - you know, so as investors, I think one of the things that we were confused by - yes, confused by a little bit is, what does this mean? I mean - but I think, you know, what was interesting, you know, obviously, Amazon was an early player with Astera as well. They have warrants at Astera Labs, and they did that.

So I guess, how do, you know - how do you think about - how should we as investors, think about these various arrangements Amazon is making with multiple players supplying the same types of products? Because - I mean, Astera is not as expanded an agreement. Astera is not going to do design on ASICs, but it does do a lot of the other things that Marvell does as well.

I'm kind of curious, when you think about this, what - maybe can you help us kind of unpack how to think about this within the broader context of how Amazon thinks about their suppliers for some of these other components maybe outside of the ASIC assets, the ASIC design products?

RS: So I think - I mean, let me give you an opinion - and let me give you a couple of facts, and then I'll give you an opinion. So it will stay in the public domain. The facts are as follows. The majority of Astera's revenues today are PCIe retimers. I mean, there are a little bit of switches. But they're trying to get into other stuff, Ethernet stuff, and cabling. But the majority - the vast majority of what they do is - comes from retimers. A vast majority of that revenue comes from AWS. So those are public facts. You can read it in the earnings releases.

The - that dependence is not necessarily, of course, as an investor, a good thing for any company. But I think there are other issues that they have to contend with, one of which is that if there was an ASIC supplier with this case, Marvell. And an ASIC supplier was also providing other components, then retimers become, oh, ok, I could get them from Astera. I could get them from the person who's also doing my ASICs.

And I don't have to worry about compatibility of those retimers. I think Astera benefited from being an early mover in that market and having some vigorous support from certain leadership at AWS. I think this is the time now for them to prove that they can stand on their own. And time will tell where that goes. I have - I don't have an opinion or any insight specifically to - will they make it the other side of the innovator's dilemma? I don't know the answer, and I don't think they do either. That's a matter of time.

I think the bigger problem they have is, there will be a time, that time maybe 12 months, 18 months, 20 months, where the revenue they derive from PCIe retimers, starts to ebb away. And if they haven't got meaningful support for other products, then they have a problem. And the issue, I think, isn't so much, is it retimer from Astera versus retimer from Marvell, or Broadcom, or anybody else?

The issue, I think is, the cadence of GPUs at least in particular, which NVIDIA has publicly stated as a yearly cadence, doesn't match the cadence of PCI. PCI 6 is getting deployed now. PCI 7 - though you can talk about PCI 7, you know, physical connectivity - and Marvell demonstrated that at OCP recently. But PCI 7, as a standard, at least a year and a half away and maybe more, which means even if you were to prefetch some of that and use the draft standard to build a retimer, you're two and a half years away from that.

So now you have a GPU market that's moving at a yearly cadence and a PCI market that's moving at a two- or three-year period cadence, and it's probably not keeping up with the performance. By their own admission, PCI 6 just about keeps up with Blackwell. Let's assume Rubin is at least 50% faster, maybe more, let's just say, conservatively 50% faster, you've got a different problem. And I think you will see a shift in the data center from the connectivity instead of being PCI in the scale-up networks to be high-speed Ethernet.

You notice Blackwell at least has a 1.6 terabit port and our friends at Marvell, my former employer, has announced earlier this year, has gone in production of 1.6T transceiver as well. So I think there's a technology shift that is much more likely to hurt Astera than - and yes, a competitor who's now close with their biggest customer is a problem, nevertheless. But the technology shift is a more meta-problem for them. Sorry, a long-winded answer.

SD: No, it was a great answer, Raj. It's super helpful. I guess, I want to unpack that a little bit because I want to make sure I understand parts of it and our investors do and our clients do as well. When you talk about the yearly versus kind of two-year cadence, you know-

RS: Three-year, yes.

SD: Three year, yes. So PCI 7 being, I don't know, two years away - maybe two and half years away, who knows. But thinking about what you just said about retimers not able to keep up with innovation.

RS: No, no, no. PCI not being able to keep up. Retimers is just an artifact of what do you do with the copper traces?

But yes. I mean, there are retimers on Ethernet as well, by the way. This is - retimers are not a new thing. They've been around for 100 years - not 100 years, but a long time.

The issue becomes as you get your data speeds high enough and the distances are long enough, you need to clean up the signal and retransmit it. So you extract the clock, you get the signal out, you reformat it and you resend it. You don't just amplify it. If you amplify it, you amplify the noise as well.

SD: Ok. I just want to make sure - so ok-

RS: The issue as a technology and how fast it can go, not so much the retimer. The retimers, they're just functioning against whatever the protocol is in the SerDes, right?

SD: Ok. That's super helpful. I appreciate that color and that explanation. So, you know, if that PCI market, I guess, it doesn't move - so I guess I'm just trying to understand exactly like you're saying people are going to move to Ethernet alternatives and that's in lieu of - I want to make sure on the fact - because I'm just not sure I - from a technical standpoint because I think it's an important point. If the PCI market can't move with the GPU innovation curve, the point is that the alternative is Ethernet, I guess, I want to make sure-

RS: Possible. Well, there's two things. One, there's Ethernet at faster speeds than currently envisaged, which currently the industry and the data centers are moving from 400 gig to 800 gig, so that's one. The second part of that is you say, oh, instead of - and to get to 800, I'm doing 100 gig per lane, ok? That gets me to 800. What Marvell did and they did it, I think with very good reason. And I suspect - I don't know, but I suspect there's an ASIC out there, at least the one hyperscaler that we recently talked about that also has a 1.6T port as well as Blackwell having a 1.6T port.

And that you do 200 gig per lane instead of 100, which is harder in a conventional Ethernet thing, and they're doing it using PAM4, which is their optical technology for interconnect, and they're layering that on top of Ethernet as a protocol, which is a great idea. The - that allows you to now transmit at 1.6T. Of course, Ethernet is a packetized protocol. So you have to consider that in your scheme of things. But it's a protocol that's very well understood and a very rich ecosystem, even more so than PCI.

So if you look up at that and you say, if I'm NVIDIA, I have NVLink. I have less reliance on the thing. And yes, I need PCI for the other side of it. But ok, I could use Ethernet. I could use some other variant of NVLink with Rubin. I don't know. At least - so I think NVIDIA is in a different position from a hyperscaler that's doing - deploying a mixture of NVIDIA and their own ASICs and their own CPUs because the connectivity now is more in question when you go beyond certain speeds.

SD: Ok. So if you think about that - so broadly speaking, you know, using the - yes, I think I follow that which is super helpful. So then what does the sector - I mean, I guess the evolutionary component of what's happening on the - for GPUs and the launch of Blackwell, where do we then - so then your point of reliance on this sector seems to be - it seems to be your point is like, this is a tough sector to rely on because of what's happening. Where do you see then - like I guess then how does this market evolve from here?

I mean, we've talked a little bit about it, but like what then ends up being the role? And like what do the Astera's and the Marvell's - so Marvell has an over - have an overarching partnership, it sounds like with AWS. So for them, it's less about one typical, one specific product. Your point about - so I guess they seem well positioned because it's like ultimately, we're going to move to custom ASICs or ASICs or whatever you want to call it.

ASICs are going to be a big part and a big theme. AWS re:Invent has been about that. But, you know, within that, how do the - is there, I guess, then a commoditization of these components? Or do these components end up kind of becoming less relevant trying to think through like what do we - how do we think about the component aspect of it, you know, whether it's PCI or say, retimers or switches? How do these things all evolve?

RS: Let's - there are several questions there, but let's unpack them one by one. I think there is - data centers have certain topologies. A Microsoft data center topology is different from an AWS topology, depending on whether they use spines or they use hubs, and then from Google as well. And then other people like Oracle help some mixture of things, smaller, but some mixture.

If you look at how all those topologies are interacted, there's today a mixture of cabling that's copper for short reaches and within the rack or inside the tray and optical when they're doing longer reaches. When you go to optical, you abbreviate the need for having retimers on Ethernet at least. And if that's the case, you sort of look up and you say, oh, why don't I do everything in optical because it's expensive.

Historically, it's been less reliable because there's more things to go wrong. I think that is getting less and less as there's more and more experience on that. But it is the case that it remains more expensive than copper. But - and certainly, for passive copper, it remains dramatically more expensive, and it's significantly more expensive at least today than even active electrical cables, ok?

So if the move becomes I'm getting to a point where I can no longer use passive copper, I move everything to active electrical cables, ok? Then maybe it's not a bridge too far to move a large portion of my data center to optical, which abbreviates the need for retimers in the first place. If I'm going to do retimers and I can't keep up with what I need to do, then - and I'm going to do to Ethernet, the same thing applies.

If you're doing long reaches, you still need Ethernet retimers as well. And Ethernet retimers have been around a long time. Marvell has them. Broadcom has them. They have gearboxes for them. We take the data rates up and down through the gearboxes. So there's a whole bunch of things. I noticed by the way, that our friends at Astera - just to circle back to that, have now introduced a 800-gig retimer for Ethernet as well. So they can - I suspect, see the same winds blowing as we're talking about in this call.

SD: But the issue is even 800, you're not at 1.6, which is what you probably-

RS: You're not. You're not. But you could take a view that I'm going to take it by 1.6T port and spread it to two 800s. So depending on how you're transmitting your data and how you're connecting things, you could manage it that way.

SD: I mean, is there a latency or performance issue with-

RS: Sure. It's actually a different thing, because you now have to - somewhere along the line, if you really want a true 1.6T pipe, you got to combine those two flows together. You do that in a NIC, you could do that in a smart switch, but there's something else has to get involved to do it, which will add latency, might also add other issues as well.

SD: Got it. Ok. So we've got - so when we think about this retimer market, it sounds like we've got a couple issues - I mean, we have this PCIe fundamental technology issue. It sounds like on the retimer side, we have this copper versus optical issue, which is what is-

RS: I'm just stating all the threats that are out there to that market. Some will happen in different points in time. Some may happen in lesser or greater extent than we envisage today, the market will decide that. And it may be that different hyperscalers adopt different approaches, and more aggressive ones, in some cases, are less aggressive than others.

So transition won't be - in a single day, you know, suddenly you're selling a million retimers an hour and the next minute, you're selling zero. That's not what's going to happen. It's going to be up and down on all the way like this, yes.

SD: Yes. But it seems like gradually though, I guess, you know, what people have been trying to think through, I think what we're all trying to figure out is when we think about some of these products, some of these technologies, are these technologies that end up moving and shifting alongside this GenAI transformation and purchasing hardware? Or are these kind of mix shift solutions and products that exist in more of a point in time that gradually end up being more legacy versus innovation over time? And it sounds like it's more the former versus the latter given the fact that ultimately-

RS: These technologies won't go away. Connectivity is-

SD: But no technologies never really go away in fairness, yes.

RS: Sorry.

SD: No technologies ever really go away, I guess, would be-

RS: Ok. So we all agree that both the compute element and the AI element inside the data center will increase in complexity, the weights and monetization of models will increase in complexity every year for the foreseeable future. And as that happens, the amount of data you have to move gets more and more. And then you say, oh, the faster I have to move the data, the more pipe I need or the faster clocking I need to get to the same thing.

There's a consortium called Ultra Ethernet that's trying to define the next generation of Ethernet to compete with InfiniBand, and that will have a bearing. There are a number of people very keen to make that into a standard and I think it will be. I think if you look out at what's going on in that marketplace, generally it's not just that the GPUs are getting faster, which they are, but they're also getting - well, I don't know getting what it was.

There's more desire to do custom ASICs for both training and particularly for inference than there has ever been before. People who went to re:Invent saw that they are on a path to go do multiple iterations of both Trainium and Inferentia in the future. And if it's - if Graviton is a road map of where they want to go, they're now saying 50% of all instances in the last year were done on Graviton or Intel. That's amazing statement, say, 10 years ago, they were at 2% or 3%, 5%.

SD: Yes. No, it makes all sense. Just thinking about that and just tying this back to Marvell for a second, when you think - the one big question mark, I think, at least among investors' minds is, were you going to see a scenario where the hyperscalers started just working directly with TSMC and just kind of focusing on internal design teams versus working with some of the challenges with these announcements.

RS: But it's true today, right? I mean, Graviton is done entirely - 99% of Graviton is done internally. They've done directly with TSMC. They do a tape-out directly. They send the GDS to database. TSMC makes the chips. They go to STATS, have it packaged. There's no Marvell or Broadcom involved in that configuration.

But other chips, ASICs - other chips are done with - in partnerships because no hyperscaler today has a big - not only would you need a big team, and you can always hire people, you can always pay them, but you would need a multidisciplinary team. You need people who know optics, people who know Ethernet. You need to go replicate a Marvell or a Broadcom in its entirety to do what they do.

SD: Right. And that - so pushing - putting them back - yes, your point is it's almost inevitable that a Broadcom and Marvell need to be at the forefront of this ASIC push or opportunity because of the resources required to do it.

RS: I think just the breadth of their experience and IP that they possess. Now could you envisage a scenario where AWS, in particular, looks up and says, yes, I could do some of this myself. Well, clearly, they don't believe they could do all of it themselves in a cost-effective manner for the next five years. Otherwise, we wouldn't have signed that agreement.

SD: Yes, of course. That's - yes.

RS: And five years is pretty much infinite in the silicon world. Who knows what happens after five years. Because five years gives you, assuming you actually started at one design a little while ago, which maybe that's premium too, that gives you three more designs. And three generations is a long time. And then you get the power of incumbency as you find with Broadcom and TPU.

SD: Right. I want to shift to a question on the Marvell and Broadcom, maybe you can answer. I mean it felt like PCIe 5, which it seemed like only Astera did, everyone skipped.

RS: I'm sorry, PCIe 5.

SD: Yes, PCIe 5. I mean the reason Astera's market share is what it is in PCIe 5 is there's no alternative for the most part.

RS: Certainly, when they came up with it, there was no alternative.

SD: Yes. I guess the question is just like it's almost - I mean, was there I guess I'm asking this in the context of this product, this component had - at least there was a perception in the market that it wasn't necessarily that interesting or that valuable or that - or worth necessarily innovating on by some very large companies.

And then ultimately, we fast forward two or three years later, and it's now considered one of the hottest things or one of the most important components. And I think your early kind of conversation around this or discussion around this in our conversation so far has been really helpful.

But I guess if we kind of think about it, obviously, Broadcom has on their conference calls publicly talked about this market being a market that they care about, whether that's a response to the Scorpio Switch product that Astera has launched or something else is unclear. But they have made this a point, and it's rare that Broadcom would talk about $1 billion product and they're a $50 billion company on an earnings call.

So I guess can we just unpack like a little bit around like what is the - when you think about PCIes, the switches and the retimers, what is the value - how valuable is this and how commoditized can this be as the competitors who didn't focus on it now do focus on it because it's at least to some degree at least in a copper framework required on the retimer side and obviously, PCIe and PCIe 6. Yes, I'm asking a lot of questions at the same time, but can you help-

Let's start from the beginning. What is it - and I touched on it a little bit, let me just reiterate. So PCIe is not a difficult technology. It's well understood. It's well documented. There's a standard, it's published. You can go read it. You want to do a PCIe controller, yes, you could figure out how to do a PCI controller.

RS: It's not that hard. PCIe 5, maybe a little harder, fine. Ok. What is a PCIe retimer, it's actually not that hard. Now if you just do a textbook, it will have high latency, it will consume a lot of power. That's not a good thing.

The innovation then comes in, hey, how do I get the latency down? How do I get the power down? Because as long as you're inside the standard, you can innovate inside the standard. And then you can communicate anything else that has PCIe. Ok, great.

The - and there are companies, Montage in China has PCIe retimers. Broadcom, of course, now does as well, as well as Marvell does. Kandou, a company in Switzerland that I'm on the board of has just started sampling their Gen 6 PCIe retimers, which is being qualified at multiple places.

It's not that there aren't retimers. It's just Astera saw an opportunity at a moment in time where others didn't. And that's they're failing. As a technology, it's not that hard. The moat, even - I've listened to Astera earnings calls, COSMOS, which they claim to be the moat and try to compare it to CUDA, it's no CUDA. I know CUDA extremely well. COSMOS is, yes, it's interesting, but it doesn't give you anything that somebody else either can't replicate or can't do better in fact.

SD: Yes. Because so when - going back to the one conversation I've been trying to have even great at articulating is what Astera has done or built across the product lines. And so I do want to switch to Scorpio for a second, and then I'll come back to Astera. The Scorpio switches seem to be resonating pretty well with customers. It seems like Broadcom has a strong hold over that market, and these switches are really strong. It seems like you're getting some adoption.

Obviously, AWS is going to use them, and it seems like other players will as well. So you potentially have a ramp in that product line as well kind of coming. And so maybe we can just kind of touch on that opportunity for them and where you think that kind of goes to the extent that you have a view on it or your thoughts around it.

RS: I think they will do very well with Scorpio as Blackwell ramps. Now my previous point of does PCIe run at a steam past Gen 6 remains valid because post Blackwell, I think the topology - unless Rubin is only 5% or 10% or 20% faster than Blackwell, which I sincerely doubt given NVIDIA's track record. If it's a multiple, even 50% faster, 60% faster than Blackwell, some other topology has to take form.

What that is, I don't know the answer yet. Maybe it's fast Ethernet, maybe it's something else. And then that's then in two years' time, Scorpio either has to embrace that new technology, whatever that happens to be or it starts ebbing away.

SD: That makes sense. And in your point, does that applies to kind of almost everything because it's - all of these products we're talking about are with a PCIe framework.

RS: Correct. I think PCI as a fabric to run things on will not go away. It's embedded in a lot of things. It as a point-to-point, a point-to-multipoint. Even CXL, which, as you know, has got a bit of a bad rap for pooling, but it does have lots of applications in memory expansion and near memory compute. That actually runs on top of PCI. It's a protocol that runs on top of PCI. So it's not going away. It's a question of what will it get used for in the future.

SD: I see. So just to back up, these TAM estimates that people are talking about on what these markets grow at for switches and for retimers and the like. Given what you're saying and maybe they're not going to - the opportunity isn't as vast as we think because ultimately, there are just - there are enough - well, maybe it is, maybe it isn't. And to your point, it's a slower move.

So no one - I'm not here to challenge whether we go from 5,000 or 1 million retimers to 0. That's not really the point. But it does seem like the growth rates of the sectors underlying these PCIe technologies, these components probably isn't necessarily as robust as the market thinks. Is that a fair assessment?

RS: I think you should segment that statement into time lines. I think if you said what happens in the next 12 months, you will actually see a rise and a significant rise in Blackwell wraps. Maybe that's 18 months. What happens beyond that is what's in doubt.

And maybe they get lucky, maybe it's two years. But there is going to be a shift away from PCIe. And whether they're on the right side of the shift, it appears, given that they just introduced an Ethernet retimer that they see it as well. They're not stupid, right? There are some very smart people there.

SD: Sure. But the retimers for Ethernet, do those - will those be used? Will those have the same adoption that-

RS: They have the same - they are used for exactly the same purpose. Now the wrench in this is if people say, hey, the way to solve this longer term, let's say the chip after Rubin is four times faster than Rubin. You may then decide, you know what, I actually can't do this with copper at all.

Even coming off the chip, I need some other co-packaged optical technology like Lightmatter to have a whole different set of paradigm to get data in and out of the chip. That's a fundamental shift in how GPUs and CPUs work today, but that would be a bombshell for everybody around them in the ecosystem.

SD: So a pure optical framework pretty much destroys this entire market.

RS: Yes. But I don't think that - I think there's a lot of ecosystem development that has to take place for that to become a reality. It could happen for some ASIC and some data center in a CSP, but I don't think it's a general statement.

SD: Yes. Fair enough. So if we look at the - to be clear, again, you've made it really - you've done a great job sharing the PCIe framework is not hard per se. So the retimer - is the switch technology hard? Is what - they are just Scorpio-

RS: It's harder.

SD: It's harder.

RS: It's harder. Absolutely. Ok. Switch technology, all switches are - well, if you talk to CPU guys, they'll tell you switches are the easiest thing to build. If you talk to switch guys, they tell you switches are the hardest thing to build, but that's ok.

I think switches that are very high speed are hard to build. PCIe switches are both easy and hard because the protocol is extremely well understood. And because PCIe is memory map, it's actually easier to build PCIe switch than it is to build Ethernet switches because you can manage the data flows much easier.

Having said that, I was looking at the Scorpio spec, the one that's public at least, and they've done a pretty good job, but it's not something that somebody else can't replicate. And I'll give you some examples. Of course, Broadcom has PCIe switches. In older generations, even relatively small companies, second, third tier companies like - what the hell that's company called - Microchip had a PCIe switch in Gen 3 or 4, I forget now.

Marvell has PCIe switches, even though it's not public because it's not public, they don't sell it as a product. So to do - they have a CXL memory expander. To do a CXL memory expander inside that CXL chip has to be a PCIe switch. So it's not like - it's not that they haven't announced it of the product. They have the technology inside another chip. They could extract that and make a PCIe switch from it. And the CXL expander is on their website. You can read it. And as I said earlier, CXL runs on top of PCIe.

So there is likely to be - if that becomes - and it's certainly for the next two, three years, it's likely to be a very good market for our friends at Astera. If somebody wanted to compete, they would have to have started a year ago, but they could come - somebody could come up with a PCIe switch that competes with them. That's not beyond the possibility.

SD: And I want to just - so I do want to talk about that for a second. So just thinking about the current road map and how to think about the fact that we are in a market right now, which obviously is moving incredibly fast. Blackwell is launching PCIe 6 framework seems to be underway. It's obviously a very good environment for these players because of the requirements and what Blackwell anticipates and what's in these designs.

If I thought about the Marvell announcement with AWS, and I just thought in the context of where I think market shares kind of look - what the market shares kind of look like in retimers and things like that. With AWS now kind of a strategic partner, does it look like - does it feel like these other partners because - I guess when you think about dependency or how these hyperscalers think about dependency on one supplier or having multiple, obviously, they probably would favor multiple suppliers because that's just - it's a good business decision to make.

It obviously keeps pricing competitive and things like that. If you thought about - but I would love your thoughts on when we think about pricing in this market, we know the volumes are going to be the volumes, can't predict that.

But if I thought about market shares and pricing in these kinds of markets, switches is going to be a market where Astera is going to be growing share because the share is negligible right now. And retimers are going to be a place where they're going to be ceding share because obviously, share is 90% right now.

RS: Because nowhere else to go. Anybody who comes into the market take share from them.

SD: Of course. That's similar on the Scorpio side for them because Broadcom has the lion's share there and has only a share to lose. So you've got these two counterbalancing forces, right? You have - you're gaining share in one market. I mean it's very likely if that's a $1 billion market, Astera on the switch side, they could have a meaningful share of that. And then on the retimer side, that market. Can we just see - any thoughts you may have just on how to think about retimers, pricing, market shares and then we can then think about Scorpio side.

RS: I obviously, I'm not privy to Astera's pricing methodology. Let's just say a - I have a general idea of what retimers help. Let's say, a retimer is somewhere between $25 and $35 for safer grads. And in large volumes, it's $20, very large volumes. And certainly, that's the price you can get a Montage retimer in China, ok? And a switch, I don't know what they sell the switch for because it's not public at least not yet, maybe next earnings call. But let's say, it's $100, $150, that sort of range. That sounds about right. Maybe it's - anyway sub-$200.

So if you're trading decrease in unit volume at a lower ASP for increase in unit volume at a higher ASP, that's not a bad deal, particularly if you can maintain your margins because you only have to trade a smaller number, right, or the higher one. And it's also the case that even if somebody is using some other retimer, they can still use your PCIe 6, interoperable, hopefully. If you've done a job right.

SD: I do. Do you think that they end up a scenario where we shares on the retimer side just pretty fast? Or are they pretty gradual? Does Astera share stay pretty strong for a very long time? I mean forget-

RS: Retimers.

SD: Yes.

RS: Yes, I think it will erode because I think a number of people will go after that. So I think they will stay strong whether they can - well, it's inevitable that they will lose some market share in such a nominal position. The question is how much? Will they remain #1? Maybe. Would they slip to #2 possible because they're concentrated on one product as a company. They're also concentrated on one customer for that one product. So they have a double vulnerability on that.

SD: Right. But what we hear a lot about is just if retimer is not an incredibly difficult technology and PCIe 6 is the standard, just that's what it is. And anybody can replicate that standard just by-

RS: Well, not anybody. You need-

SD: Sorry. I apologize. You're right. I should have been more clear about that. Anyone of significance in the sector with an existing team can replicate that technology as what I should have said. If that turns out to be true, I guess the question just becomes like is there a scenario where like you go - I mean - and I really want to understand just from a more - can you go from 90-plus %, 100% market share in 1% to 50% in the quarter? Like I guess if I was thinking as a hyperscaler as these customers, and obviously, you've never worked with them-

RS: You can't do it in a quarter. You might do it in a year, a year and a half, you can't - it won't happen that quarter.

SD: Got it. So this is a - it's a gradual you'd lose maybe 10 points of share per quarter-

RS: But the question becomes, can you fill in that gap with Scorpio?

SD: Sure. Agreed. That is absolutely - from a - yes, I think - 100% agree with that. I was more just thinking just how should we think about the erosion of market share on their Aries line over the next year? It sounds like it's a gradual one. You're not going to see it happen overnight, which obviously makes sense.

Does price - and can I ask, does pricing here start playing a role now that there's multiple players, meaning, you know, we've heard about 15%, 20% price increases on the Aries line - sorry, for PCIe 6 versus PCIe 5. If I thought about like other competitors, Marvell, Broadcom, or whoever, you know, you obviously have the Chinese guys with PCIe retimers and switches.

They can't really penetrate the US market for obviously geopolitical issues. Does pricing - do you see the same type of pricing lift even if there's more competition? Or does - I mean, does everyone just benefit from the pricing? I'm curious.

RS: So if you fast forward this a year or two from now, you should expect Gen 6 pricing to be equal to Gen 5 pricing. Gen 5 pricing today will - Gen 6 pricing in 18 months is likely to be what Gen 5 pricing is today. They always normalize.

SD: Ok. So it will start off higher, whatever the number is and then erode.

RS: Always. Yes.

SD: Ok. Despite the fact that there's multiple players. Because if I was Broadcom and Marvell, I would always just be - whatever, if I was another competitor and I see a guy with 100% market share, I was - like how aggressively would I want to price to potentially take some of that share?

RS: Well, you price aggressively, you go do deals that involve lots of other things that PCIe retimers are part of the deal. They just do that.

SD: Right. That's the announcement on Monday. So that adds up. Ok. That's logical. And then Scorpio is the Scorpio, we'll have to see what happens. But it obviously seems like they have a very good product, and it looks like it's gaining market share and traction with players.

You know, thinking back to just Marvell for a second and tying back to that, when you think about Marvell here and their partnership with AWS, I mean, I guess, is this - is it logical to just expect all these hyperscalers and large companies to pursue partnerships like this with - you know, there's really just two players, it feels like Broadcom and Marvell. How do you think about these kinds of kind of expanded partnerships and whether you kind of expect them to continue to happen for these - for the two, I guess, major guys?

RS: You know, the AWS relationship with Marvell goes back a very long way. Way to - prior to the acquisition of Annapurna, when we were at Cavium, we built the first SmartNIC for them using ARM-based processor. I was there. So it's not new. That's 12 years old. So it's a long time, and a lot of effort by some very smart enabled people to get to this point. They've done similar things with other customers.

They did a deal five years ago for generation - three generations of baseband and processor silicon and switches for Nokia which at the time was the biggest deal they've ever done was multiple billions of dollars. This is echoes of the same deal. We'll give you five years. We'll give you dedicated teams. We'll give you, you know, first access to new technologies. You give us multiple generations, you give us some amount of market share.

And then there's trigger points on making sure that each generation, Marvell remains not just honest, but also competitive on technology when there are various ways to test that. That's the kind of thing that they would do. Would they do it again? Would they do it with Google or with Microsoft or somebody else? Maybe.

But I think those - both those relationships - Microsoft is a deeper relationship. Google is a good relationship, and it's very good on the optical side. Their first foray into the compute element has been Acxiom, which is from all accounts, done very well. So maybe that's a good starting point for a future relationship.

SD: Yes. You know, I think I also wonder just part of this, and I'm curious, you know, when you [INAUDIBLE 53:59-54:01] success in Scorpio, and it seems like Broadcom ASIC is just the level of - and feel free to comment on this if they're not. There seems like there's a decent amount of distaste for Broadcom despite their scale and their scope.

RS: So I was in the first meeting that Matt Murphy had with Andy Jassy just as he was moving away from AWS and becoming CEO. And his comment, and was there, too, was anybody but Broadcom.

SD: Oh, that sounds a lot.

RS: Yes. And I think there was a distaste prepandemic, and I think how people behaved during the pandemic colored a lot of the narrative going forward, both good and bad.

SD: What was the good, can I ask?

RS: At Marvell, we took a view that hyperscalers were sacrosanct and that we would drop Peter to pay Paul and keep hyperscalers whole. Hyperscalers and automotive were the two sectors that did not suffer and one telco client because those things become critical to the infrastructure.

And if you don't have those, then - now to the extent that prices went up from TSMC or from the OSATs or substrates, then there was an agreement that said, hey, we'll pass along actual price increase, and we'll take margin hits, but we can't just absorb all of it. But in some cases, we did absorb some. And that also changes the narrative, right? It says to people, hey, this is a partner, not just a vendor.

SD: Yes. Ok. I mean, we're almost in the-

RS: I mean, there a companies that are entirely - there are customers that are very transactional, and then you behave transactionally. That's fine.

SD: Right. Well, we're near the end of our time. Is there anything I should have asked that I didn't?

RS: No, I think you've covered a lot. I mean, the only thing you could touch on is one of the things in that announcement was switches. And I think you know that Tomahawk enjoys a very large market share for top of the rack switch. But the acquisition of Innovium gave Marvell two years ago or three years ago, no, two and a half years ago, give them a double-digit footprint on top of the rack switch at AWS. And I think that this agreement will potentially further that more.

SD: Got it. Great. So it sounds like you're pretty constructive on the entire space, but obviously wary of the competitive landscape and how it may change. And I mean, is that a fair assessment of what you think going forward?

RS: Yes. I would say so. I think technologies change over time. And as they change, there are winners and losers. If you just looked at Ethernet, when people went from 100 gig to 400 gig, there were companies that couldn't make the transition. As people are going from 400 gig to 800 gig, there are companies that will fall away.

And the same thing as they go beyond that. The transition to - from copper to optical and it will eventually happen, will also - there will be winners and losers in that. The same thing with PCI. You know, there will be winners and losers between transitions. And transitions are always dangerous because it's not clear that the same people win with each transition. And it creates an opening for competitors to come in and take market share.

SD: Got it. Raj, this is an incredible call. Thank you so much for all the insights, all the time. Really appreciate it. Let's catch up again hopefully really soon.

RS: Thank you. Look forward to it. Take care.

SD: Pleasure. Thanks again.

RS: Thank you. Bye-bye.